

The constants K and λ in Ramanujan's identity (9.2.5) of the Lost Notebook

Otakhon U. Kenjaev

Independent researcher, Khorezm, Uzbekistan

ORCID 0009-0009-3566-9285

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Abstract

On pages 270–271 of the Lost Notebook, Ramanujan stated an identity for the alternating divisor series $\sum (-1)^{n+1} \sigma_{s-1}(n) e^{-n\pi\sqrt{3}}$ whose right-hand side contains two undefined quantities K and λ . Andrews and Berndt recorded it as (9.2.5) and wrote that they were unable either to identify K, λ or to obtain an identity of Ramanujan's form. We show that the identity holds for every $\operatorname{Re} s > 2$ with $K = \mu - \nu$, $\lambda = \mu + \nu$, the sum running over coprime $\mu, \nu \geq 1$. The proof groups the Andrews–Berndt lattice sum by the six units of the Eisenstein integers. As consequences we obtain a fast evaluation of the hexagonal angular lattice sum $R(s)$ and the closed forms $R(6k) = (-1)^k (c_k E_6(\rho)^k - 3)/3$ for all $k \geq 1$. We also report randomized numerical checks.

1 Introduction

Let $\sigma_{s-1}(n) = \sum_{d|n} d^{s-1}$ and

$$L(s) = \sum_{n \geq 1} (-1)^{n+1} \sigma_{s-1}(n) e^{-n\pi\sqrt{3}}. \quad (1)$$

In Section 18 of the manuscript printed on pages 270–271 of the Lost Notebook [1], Ramanujan wrote an identity of the form

$$L(s) = -\frac{2\Gamma(s)\zeta(s)}{(2\pi)^s} \left\{ \cos \frac{\pi s}{6} + 2 \cos \frac{\pi(s+1)}{6} \cos \frac{\pi(s-1)}{6} \sum \frac{\cos(s \tan^{-1} \frac{K}{\lambda\sqrt{3}})}{(\mu^2 + \mu\nu + \nu^2)^{s/2}} \right\}, \quad (2)$$

recorded by Andrews and Berndt as (9.2.5) in [2, §9.2]. After (9.3.2) they write that K and λ are undefined and that they are “unable either to identify them or to obtain an identity of the form given by Ramanujan”. They prove instead an identity for $L(s)$ as a sum over the half-plane $\mu \geq 1$, $\nu \in \mathbb{Z}$ (Section 2).

Contribution. (i) Theorem 1: (2) holds with $K = \mu - \nu$, $\lambda = \mu + \nu$, $\gcd(\mu, \nu) = 1$, $\mu, \nu \geq 1$, for $\operatorname{Re} s > 2$. (ii) Theorem 2: the slowly convergent sum $R(s)$ in (3) is expressed through the rapidly convergent series (1). (iii) Corollary 1: closed forms of $R(6k)$ for all k . We do not claim new methods: at integer s the identity is the orbit decomposition of the classical lattice sum for the Eisenstein series $E_s(\rho)$ (Remark 3). The new point is the identification of Ramanujan's K, λ and its consequences for real and complex s .

2 Notation and the Andrews–Berndt identity

Let $\omega = e^{i\pi/3}$, so $\omega^2 = \omega - 1$, and let $\rho = e^{2\pi i/3}$. For $w = a + b\omega \in \mathbb{Z}[\omega]$ put $N(w) = |w|^2 = a^2 + ab + b^2$; w is *primitive* if $\gcd(a, b) = 1$. The units are $\pm 1, \pm\omega, \pm\omega^2$. Write

$$R(s) = \sum_{\substack{a, b \geq 1 \\ \gcd(a, b) = 1}} \frac{\cos(s\psi_w)}{N(w)^{s/2}}, \quad \psi_w = \arg w - \frac{\pi}{6} \in \left(-\frac{\pi}{6}, \frac{\pi}{6}\right), \quad \tan \psi_w = \frac{b - a}{\sqrt{3}(a + b)}. \quad (3)$$

The series converges absolutely for $\operatorname{Re} s > 2$. Andrews and Berndt [2, p. 217] prove that, for $\operatorname{Re} s > 2$,

$$L(s) = -\frac{\Gamma(s)\zeta(s)}{(2\pi)^s} \sum_{\substack{\mu \geq 1, \nu \in \mathbb{Z} \\ \gcd(\mu, \nu) = 1}} \frac{\cos(s \arg z)}{|z|^s}, \quad z = e^{i\pi/6}(\mu + \nu\omega). \quad (4)$$

Here $|z|^2 = \mu^2 + \mu\nu + \nu^2$ and $\tan(\arg z) = (\mu + 2\nu)/(\sqrt{3}\mu)$ with $|\arg z| < \pi/2$.

3 Two lemmas

Lemma 1 (orbit partition). *Let $H = \{\mu + \nu\omega : \mu \geq 1, \nu \in \mathbb{Z}\}$. If w is primitive and not a unit, exactly three of its six associates lie in H , namely $w, \omega^{-1}w, \omega^{-2}w$ when $w = a + b\omega$ with $a, b \geq 1$; moreover every such orbit contains exactly one element with $a, b \geq 1$. For the unit orbit exactly two associates, 1 and ω^{-1} , lie in H , and $\pm\omega$ lie on its boundary.*

Proof. Write $\mu + \nu\omega = x + iy$. Then $\mu = x - y/\sqrt{3}$, so for integers $\mu \geq 1 \iff \mu > 0 \iff \arg(\mu + \nu\omega) \in (-2\pi/3, \pi/3)$, an open sector of width π . Multiplication by ω adds $\pi/3$ to the argument, so the six associates of w have arguments $\theta + j\pi/3$ ($0 \leq j \leq 5$). If $\theta \notin (\pi/3)\mathbb{Z}$, exactly three of them fall in an open sector of width π , and exactly one falls in the open sector $(0, \pi/3)$, which is the set $a, b \geq 1$. If $\theta \in (\pi/3)\mathbb{Z}$, then $w = tu$ with $t > 0$ and u a unit; primitivity forces $t = 1$. The associates of 1 have arguments $0, \pm\pi/3, \pm 2\pi/3, \pi$; the open sector contains 0 and $-\pi/3$ (the elements 1 and ω^{-1}), and $\pm\omega$ lie on its boundary rays $\mu = 0$. Units act by matrices in $\operatorname{GL}_2(\mathbb{Z})$ on (a, b) , hence preserve primitivity. $\square$

Lemma 2 (sign). $\cos(s \tan^{-1}(-x)) = \cos(s \tan^{-1} x)$. Hence in (2) the choices $K = \mu - \nu$ and $K = \nu - \mu$ give the same sum; the set $\{a, b \geq 1, \gcd = 1\}$ and $N(w)$ are also symmetric under $a \leftrightarrow b$.

4 Main results

Theorem 1. *For $\operatorname{Re} s > 2$, identity (2) holds with $K = \mu - \nu$, $\lambda = \mu + \nu$ and the sum over coprime $\mu, \nu \geq 1$; equivalently*

$$L(s) = -\frac{2\Gamma(s)\zeta(s)}{(2\pi)^s} \left\{ \cos \frac{\pi s}{6} + 2 \cos \frac{\pi(s+1)}{6} \cos \frac{\pi(s-1)}{6} R(s) \right\}.$$

Proof. By absolute convergence we may regroup (4) by unit orbits (Lemma 1). For $w = a + b\omega$ with $a, b \geq 1$ the three associates in H have $\arg z = \psi_w + \pi/3, \psi_w, \psi_w - \pi/3$ (since $z = e^{i\pi/6}w$), and

$$\cos\left(s\left(\psi + \frac{\pi}{3}\right)\right) + \cos(s\psi) + \cos\left(s\left(\psi - \frac{\pi}{3}\right)\right) = \cos(s\psi)\left(1 + 2 \cos \frac{\pi s}{3}\right) = 4 \cos \frac{\pi(s+1)}{6} \cos \frac{\pi(s-1)}{6} \cos(s\psi).$$

The unit orbit contributes $\cos(\pi s/6) + \cos(-\pi s/6) = 2\cos(\pi s/6)$. Substituting into (4) gives the claim, and $\tan \psi_w = (b-a)/(\sqrt{3}(a+b))$ with Lemma 2 gives $K = \mu - \nu$, $\lambda = \mu + \nu$ (with $a = \mu$, $b = \nu$). $\square$

Theorem 2 (fast evaluation). *If $\operatorname{Re} s > 2$ and $s \not\equiv 2, 4 \pmod{6}$, then*

$$R(s) = \frac{-\frac{L(s)(2\pi)^s}{2\Gamma(s)\zeta(s)} - \cos \frac{\pi s}{6}}{2 \cos \frac{\pi(s+1)}{6} \cos \frac{\pi(s-1)}{6}}.$$

Since $e^{-\pi\sqrt{3}} \approx 0.0043$, about twenty terms of (1) give double precision, while the lattice sum (3) truncated at $N(w) \leq X^2$ has error of order X^{2-s} .

Remark 1 (degenerate values). For $s \equiv 2, 4 \pmod{6}$ the coefficient of $R(s)$ vanishes and Theorem 1 carries no information on $R(s)$. For even integers s in these classes one has $E_s(\rho) = 0$, because every monomial $E_4^i E_6^j$ of weight s then contains E_4 and $E_4(\rho) = 0$.

Corollary 1 (closed forms). *For every integer $k \geq 1$,*

$$R(6k) = (-1)^k \frac{E_{6k}(\rho) - 3}{3}, \quad E_{6k}(\rho) = c_k E_6(\rho)^k, \quad E_6(\rho) = \frac{27 \Gamma(1/3)^{18}}{512 \pi^{12}},$$

where $c_k \in \mathbb{Q}$ is the coefficient of E_6^k when E_{6k} is written as a polynomial in E_4, E_6 . The first values are $c_1 = 1$, $c_2 = \frac{250}{691}$, $c_3 = \frac{5500}{43867}$, $c_4 = \frac{10285000}{236364091}$, $c_5 = \frac{26021050000}{1723168255201}$. In particular

$$R(6) = 1 - \frac{9 \Gamma(1/3)^{18}}{512 \pi^{12}}, \quad R(12) = \frac{30375 \Gamma(1/3)^{36}}{90570752 \pi^{24}} - 1.$$

Proof. With $q = e^{2\pi i \rho} = -e^{-\pi\sqrt{3}}$ we have $L(s) = -\sum \sigma_{s-1}(n)q^n = \frac{B_s}{2s}(E_s(\rho) - 1)$ for even s . At $s = 6k$: $\cos(\pi k) = (-1)^k$, $2 \cos \frac{\pi(6k+1)}{6} \cos \frac{\pi(6k-1)}{6} = \frac{3}{2}$, and $\zeta(6k) = (-1)^{3k+1} B_{6k} (2\pi)^{6k} / (2(6k)!)$, which give $\frac{L(6k)(2\pi)^{6k}}{2\Gamma(6k)\zeta(6k)} = (-1)^{k+1} \frac{E_{6k}(\rho) - 1}{2}$. Theorem 2 then gives the formula for $R(6k)$. The value of $E_6(\rho)$ is the Chowla–Selberg evaluation [3], and $E_4(\rho) = 0$ kills every monomial containing E_4 . $\square$

Remark 2 (consistency). As $k \rightarrow \infty$, $E_{6k}(\rho) \rightarrow 3$ (the six units contribute $u^{-6k} = 1$ each, halved), so $R(6k) \rightarrow 0$, in agreement with $R(s) \sim 3^{-s/2}$ from the smallest term $w = 1 + \omega$. Numerically

$$E_{36}(\rho) = 3.00000000774\dots, \quad E_{60}(\rho) = 3.00000000000001\dots$$

Remark 3 (what is classical). For integer s , (4) is the Lipschitz (Fourier) expansion of the lattice sum defining $E_s(\rho)$, and Corollary 1 is its sector form. Ramanujan’s (9.2.5), however, is stated for real s , where no modular interpretation of $L(s)$ is available; Theorem 1 supplies the missing K, λ there, and Theorem 2 holds for all complex s with $\operatorname{Re} s > 2$ outside the degenerate classes.

5 Numerical evidence

All computations are reproducible with the scripts archived at [4] (Python, NumPy, mpmath, SymPy). They are empirical checks in double or multiple precision, not rigorous error bounds.

- *Lemma 1.* All 35088 primitive w with $|a|, |b| \leq 120$: no violation of “exactly three”; unit orbit: two in H , boundary $\{\pm\omega\}$.

- *Fixed s .* At $s = 4.7, 5.5, 6, 7.5, 8.7, 9.25, 13.1$ the two sides of Theorem 1 agree to relative $8 \cdot 10^{-16}$ – $3.5 \cdot 10^{-12}$ (3.8 million coprime pairs, left side to 30 digits); other natural choices of K, λ miss by $5 \cdot 10^{-6}$ – $2 \cdot 10^{-1}$ (at $s = 6$ the choice $K = \mu + 2\nu$, $\lambda = \mu$ also agrees, since its angles differ from ψ_w by multiples of $\pi/3$). The script also includes the degenerate values $s = 4, 10$, where every choice agrees.
- *Random s .* For 40 values $s \sim U(2.1, 20)$ the lattice sum was truncated at $|w| \leq 2400$ with the tail correction $T(X) = \frac{12}{\sqrt{3}\pi^2} \cdot \frac{2\sin(\pi s/6)}{s} \cdot \frac{X^{2-s}}{s-2}$; in all 40/40 cases the relative discrepancy was at most ten times the estimated error $|R_X - R_{X/2}|$ (plus 10^{-13}); the control $K = \mu - \nu$, $\lambda = \mu$ never came closer than $2.3 \cdot 10^{-9}$.
- *Complex s .* $s = 5 + 3i$: $2.0 \cdot 10^{-15}$; $s = 7.3 - 2i$: $5.4 \cdot 10^{-17}$; $s = 3.5 + i$: $5.8 \cdot 10^{-11}$.
- *Corollary 1.* For $k \leq 5$ the closed form agrees with Theorem 2 to $2 \cdot 10^{-40}$ (40 digits); see Table 1.

k	c_k	$R(6k)$
1	1	0.039486299736351459...
2	250/691	0.0013602406593698314...
3	5500/43867	$5.0757704420300927 \dots \times 10^{-5}$
4	10285000/236364091	$1.88165499540916937 \dots \times 10^{-6}$
5	26021050000/1723168255201	$6.96920718649738465 \dots \times 10^{-8}$

Table 1: Closed forms of Corollary 1, checked against Theorem 2.

References

- [1] S. Ramanujan, *The Lost Notebook and Other Unpublished Papers*, Narosa, New Delhi, 1988.
- [2] G. E. Andrews and B. C. Berndt, *Ramanujan's Lost Notebook, Part IV*, Springer, New York, 2013, Chapter 9.
- [3] S. Chowla and A. Selberg, On Epstein's zeta-function, *J. Reine Angew. Math.* 227 (1967), 86–110.
- [4] O. U. Kenjaev, The constants K and λ in Ramanujan's identity (9.2.5) of the Lost Notebook, with a hexagonal lattice sum calculator, Zenodo, 2026, doi:10.5281/zenodo.22987480.